

1. Simulation process. Initialize with a random distribution of a pair of average concentration and internal energy $(\bar{c}, \bar{E})$.

2. Hyperbolic Equation System. In the simulatin of partial melting system, there arises multidimensional hyperbolic equations systems.

$$\mathbf{U}_t + \nabla \cdot [F(\mathbf{U}) - A(\mathbf{U}\nabla\mathbf{U})] = 0$$

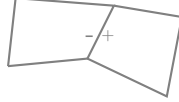
$\mathbf{U}$ consists of two variables, average concentration of one species $\bar{c}_1$ and average total internal energy $\bar{E}$.

$$\bar{c}_t + \nabla \cdot (\mathbf{v}_e \bar{c} - \kappa D \nabla \bar{c}) = 0$$

The finite volume discretization can be written as:

empty

where $\kappa = \frac{\phi_f c_f}{\bar{c}}$ need definition on edge shared between two cells.



There are two κ values defined on both sides of the shared edge.

$$\kappa^\pm = \frac{\phi_f c_f^\pm}{\bar{c}^\pm}$$

where ϕ_f has definition on the edge.

3. Eutectic phase behavior. The binary eutectic phase behavior describes the chemical behavior of two immiscible crystals from a completely miscible solution.

On a typical eutectic phase behavior diagram, we can locate any phase situation using temperature and composition information. We follow phase behaviors details described in [?]. We made reasonable adjustments for our mantle system.

There are few key elements that allow precise calculation of phase region and corresponding volumetric fractions.

Several important points

- Mass composition of solid phase 2 in liquid phase χ only relates to temperature.
- Melting temperature related to pressure. Pressure release is one of the main reasons why melting happens in mantle.
- Eutectic composition is a function of temperature and pressure. $\chi_e(\mathbf{T}, \mathbf{P})$

3.1. Detailed description of phase behavior. In mantle, we mostly have olivine and orthopyroxene (opx) mixture in the rock. Like eutectic melting system e.g. salt and ice, the presence of one phase would largely lower the melting temperature of another phase to the eutectic melting point.

3.1.1. Region I. The conditions to be in region I are

$$(3.1) \quad \mathbf{C} = 0 \quad \mathbf{H} \leq \mathbf{H} = 1$$